Combining Epigenome and Genetic Mutation Data from GWAS for the Study of Disease-Relevant Cell Types

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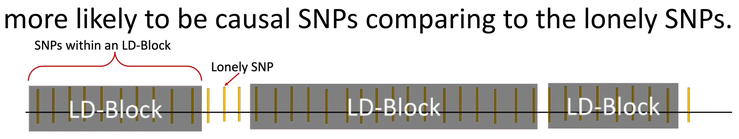
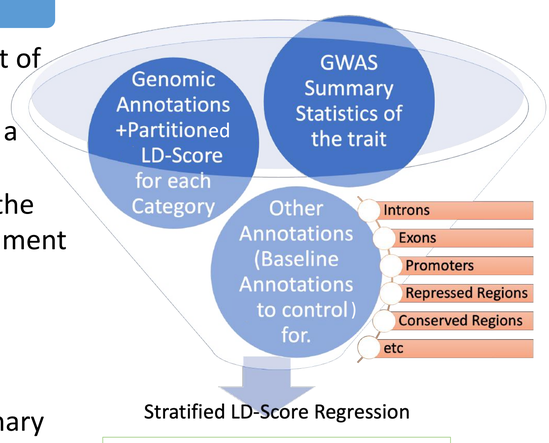
Introduction

- Genome Wide Association Studies (GWAS) Summary Statistics alone fail to compute the genetic heritability of a certain trait, since it does not take into account the Linkage Disequilibrium (LD) between SNPs within the LD-Block.
- The probability of the causal SNP of a certain phenotype is directly proportional to its LD. In other words, the SNPs located in the LD-block are more likely to be causal SNPs comparing to the lonely SNPs.



Methods

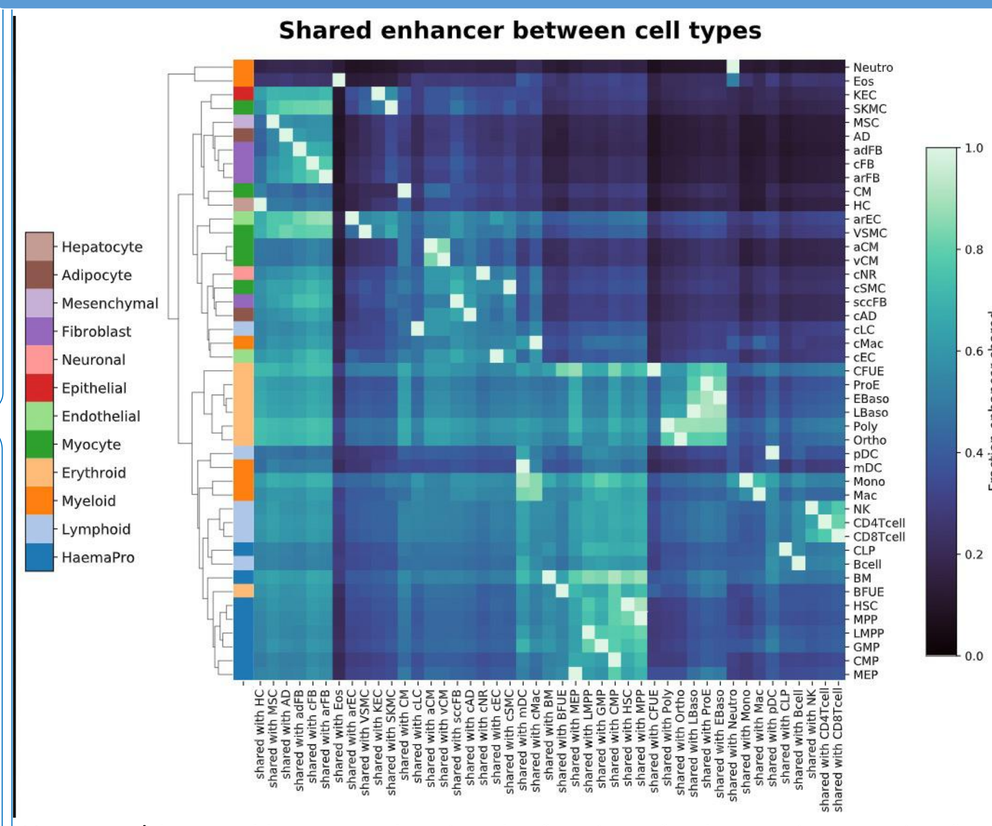
- In this work, the heritability enrichment of various enhancers extracted from 45 different cell types are estimated using a method called "Stratified LD-Score Regression"[8] (S-LDSC). Where the LD is the main indicator of the heritability enrichment of a certain functional category.
- We analysed 66 different functional categories (45 enhancers + 21 baseline category) against 7 sets of GWAS summary statistics of the phenotypes related to some cardiovascular common conditions. The functional categories and the conditions are described below in this poster.

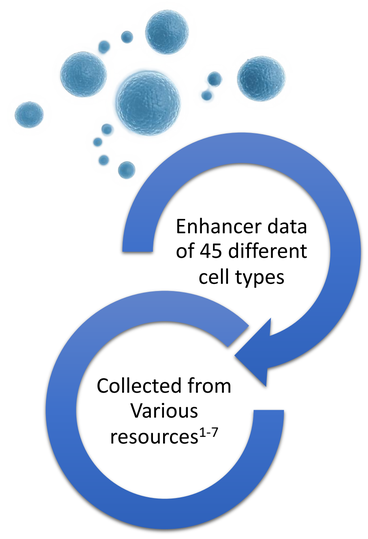


The heritabilty enrichment of a Category of an annotation =

$$= \frac{Proportion\ of\ SNP\ heritability\ in\ Category\ ci}{Proportion\ of\ SNPs\ in\ Category\ ci} = \frac{h^2(C)/h^2}{|C_i|/M}$$

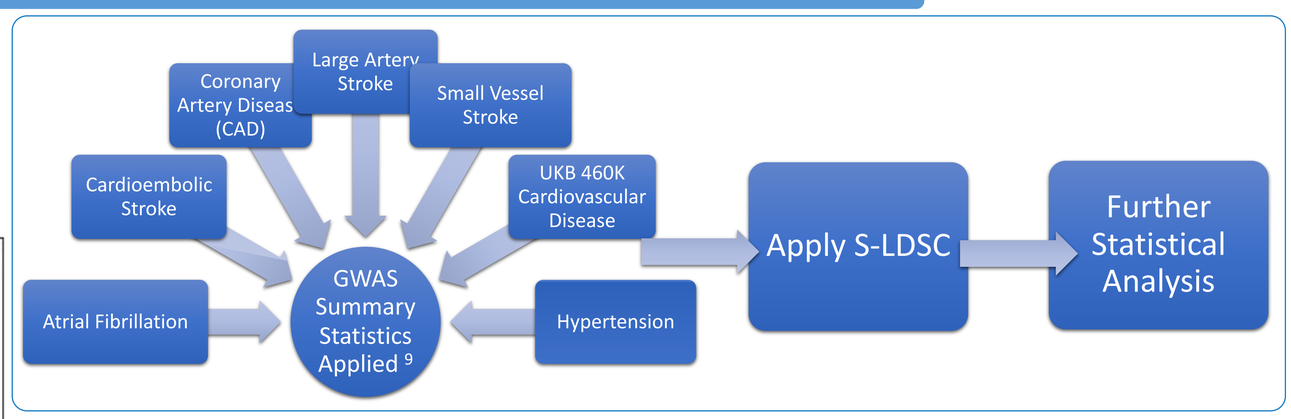
Data Collection and Acquisition





Heatmap/clustered heatmap showing the fraction of enhancer intersection for a collection of 45 cell types. Enhancers were annotated by ATAC-seq, DNase-seq and H3K27ac ChIP-seq. For more details check the references [1-7]

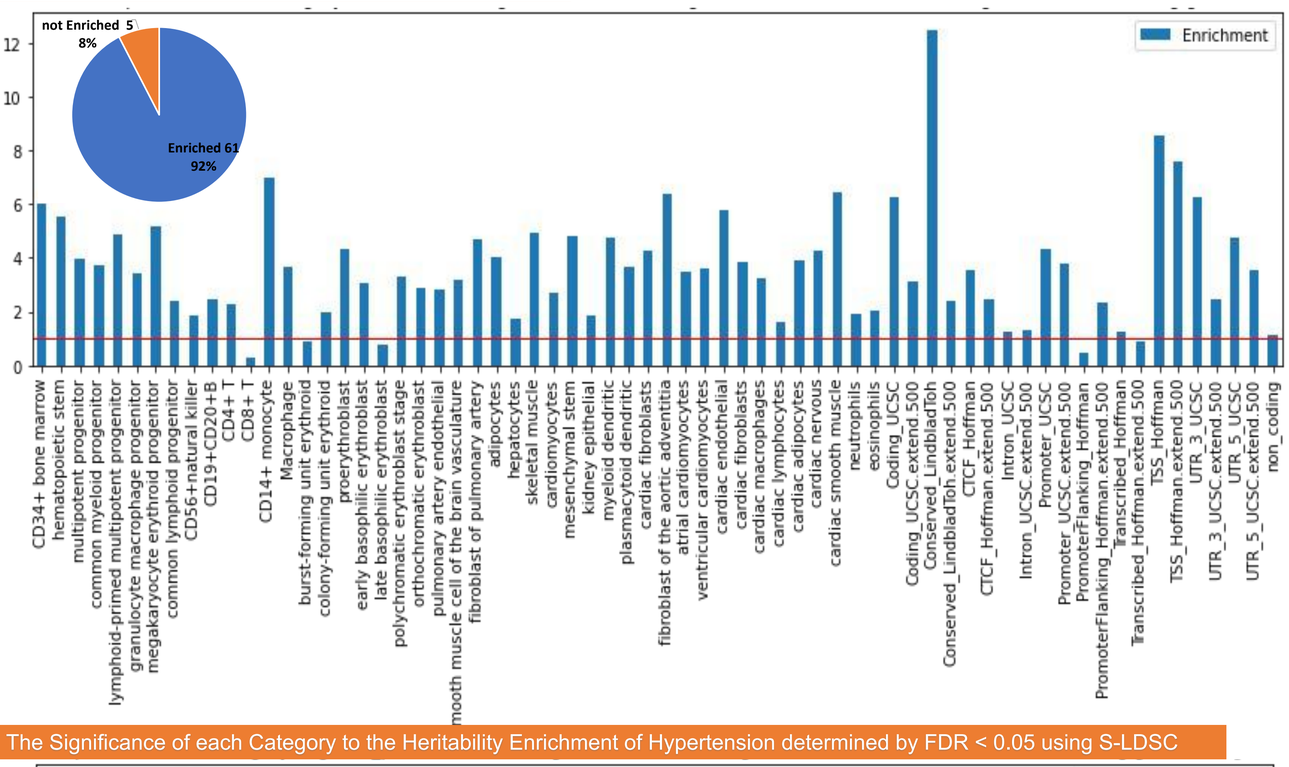
Experimental Results



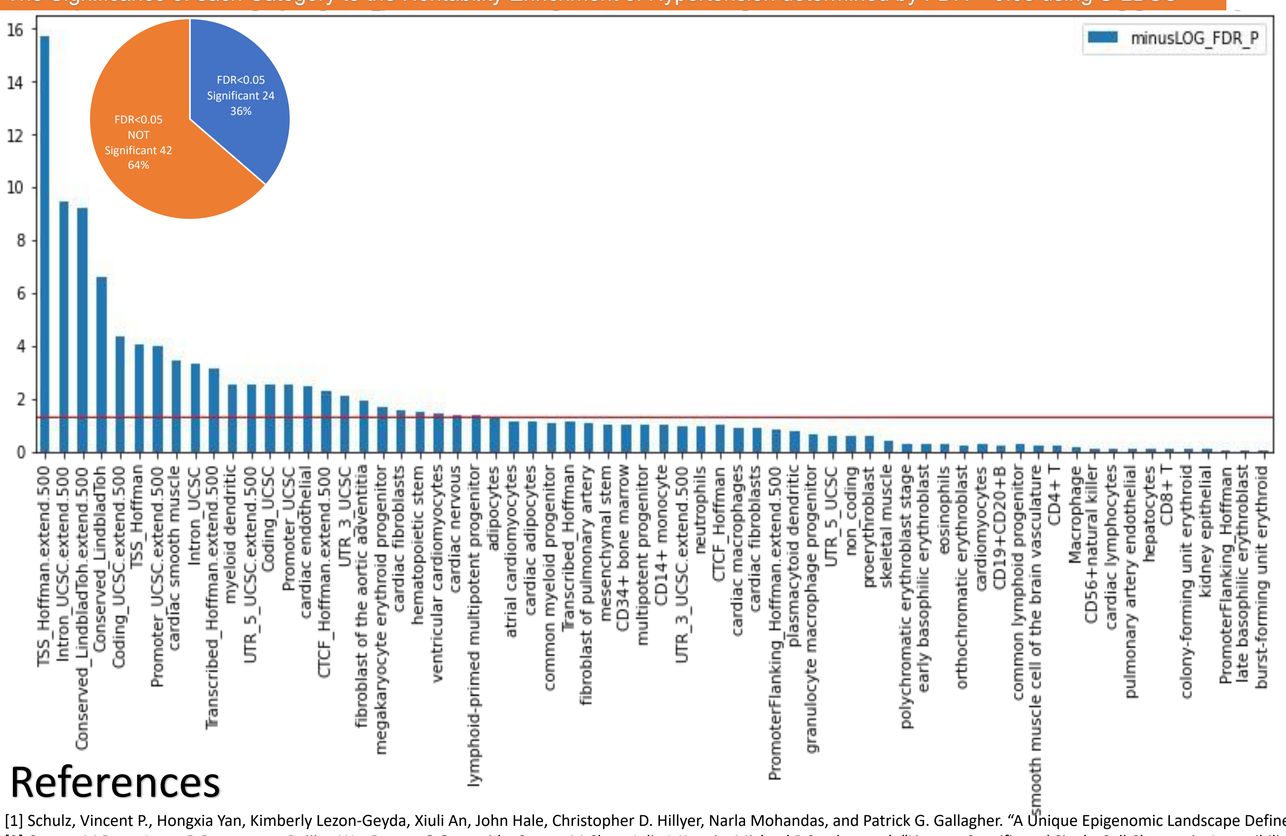
- S-LDSC was applied to each GWAS separately to compute the enrichment and the P-value of the enrichment for each functional category. The –log(P) for each category corresponding to the used GWAS is shown below. The regions with darker red indicates that this functional category shows high heritability enrichment when studying the trait represented in the GWAS.

Hypertension Focused Experimental Results

Hypertension Enrichment Values Using S-LDSC



The Significance of each Category to the Heritability Enrichment of Hypertension determined by FDR < 0.05 using S-LDSC



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